Max-Share Misidentification

Online Appendix

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A Proofs

A.1 Preliminary Results

Lemma 1

Proof. The lemma follows from:

$$\delta_i' \left[\sum_{h \in \mathcal{H}} \widetilde{\Psi}_h \theta \theta' \widetilde{\Psi}_h' \right] \delta_i = \sum_{h \in \mathcal{H}} \operatorname{Tr} \left(\theta' \widetilde{\Psi}_h' \delta_i \delta_i' \widetilde{\Psi}_h \theta \right) = \theta' \left[\sum_{h \in \mathcal{H}} \widetilde{\Psi}_h' \delta_i \delta_i' \widetilde{\Psi}_h \right] \theta.$$

To see this, note that the objective is a scalar and thus equals its own trace. Because the trace is a linear mapping, the objective becomes $\sum_{h \in \mathcal{H}} \operatorname{Tr} \left(\delta_1' \widetilde{\Psi}_h \theta \theta' \widetilde{\Psi}_h' \delta_1 \right)$. The first equality then follows since $\operatorname{Tr} \left(\delta_1' \widetilde{\Psi}_h \theta \theta' \widetilde{\Psi}_h' \delta_1 \right) = \operatorname{Tr} \left(\theta' \widetilde{\Psi}_h' \delta_1 \delta_1' \widetilde{\Psi}_h \theta \right)$ for each h. The second equality holds again by the linearity of the trace operator. □

Inner Product in the Frequency Domain Problem

Lemma 2. *$\langle \psi_j, \psi_{j'} \rangle^{freq} \equiv \int_{\omega \in \Omega} \Gamma_{1j}^{\text{Re}}(\omega) \Gamma_{1j'}^{\text{Re}}(\omega) + \Gamma_{1j}^{\text{Im}}(\omega) \Gamma_{1j'}^{\text{Im}}(\omega) d\omega$ is an inner product mapping $\mathbb{R}^\infty \times \mathbb{R}^\infty \to \mathbb{R}$.*

Proof. Conjugate symmetry is satisfied since:

$$\begin{aligned} \langle \psi_{j_1}, \psi_{j_2} \rangle^{freq} &= \overline{\int_{\omega \in \Omega} \Gamma_{1j_1}^{\text{Re}}(\omega) \Gamma_{1j_2}^{\text{Re}}(\omega) + \Gamma_{1j_1}^{\text{Im}}(\omega) \Gamma_{1j_2}^{\text{Im}}(\omega) d\omega} \\ &= \int_{\omega \in \Omega} \Gamma_{1j_2}^{\text{Re}}(\omega) \Gamma_{1j_1}^{\text{Re}}(\omega) + \Gamma_{1j_2}^{\text{Im}}(\omega) \Gamma_{1j_1}^{\text{Im}}(\omega) d\omega = \overline{\langle \psi_{j_2}, \psi_{j_1} \rangle^{freq}}. \end{aligned}$$

Linearity is satisfied since, denoting the h-th element in ψ_j by $\psi_{h,j}$ for $h \in \mathbb{Z}_{\geq 0}$, we have:

$$
\begin{aligned}
&\langle a\psi_{j_1} + b\psi_{j_2}, \psi_{j_3}\rangle^{freq} \\
&= \int_{\omega\in\Omega} \left[\sum_{h=0}^{\infty} (a\psi_{h,j_1} + b\psi_{h,j_2}) \cos(\omega h)\right] \left[\sum_{h=0}^{\infty} \psi_{h,j_3} \cos(\omega h)\right] \\
&\qquad\quad + \left[-\sum_{h=0}^{\infty} (a\psi_{h,j_1} + b\psi_{h,j_2}) \sin(\omega h)\right] \left[-\sum_{h=0}^{\infty} \psi_{h,j_3} \sin(\omega h)\right] d\omega \\
&= a\Big\{\int_{\omega\in\Omega} \left[\sum_{h=0}^{\infty} \psi_{h,j_1} \cos(\omega h)\right] \left[\sum_{h=0}^{\infty} \psi_{h,j_3} \cos(\omega h)\right] \\
&\qquad\quad + \left[\sum_{h=0}^{\infty} \psi_{h,j_1} \sin(\omega h)\right] \left[\sum_{h=0}^{\infty} \psi_{h,j_3} \sin(\omega h)\right] d\omega\Big\} \\
&\quad + b\Big\{\int_{\omega\in\Omega} \left[\sum_{h=0}^{\infty} \psi_{h,j_2} \cos(\omega h)\right] \left[\sum_{h=0}^{\infty} \psi_{h,j_3} \cos(\omega h)\right] \\
&\qquad\quad + \left[\sum_{h=0}^{\infty} \psi_{h,j_2} \sin(\omega h)\right] \left[\sum_{h=0}^{\infty} \psi_{h,j_3} \sin(\omega h)\right] d\omega\Big\} \\
&= a\langle \psi_{j_1}, \psi_{j_3}\rangle^{freq} + b\langle \psi_{j_2}, \psi_{j_3}\rangle^{freq},
\end{aligned}
$$

where the second equality follows from the linearity of the integral and summation operators and a rearrangement of terms. Positive-definiteness follow because for non-zero (over $\mathfrak{H}$) ψ_{j_1},

$$
\langle \psi_{j_1}, \psi_{j_1}\rangle^{freq} = \int_{\omega\in\Omega} \left[\sum_{h=0}^{\infty} \psi_{h,j_1} \cos(\omega h)\right]^2 + \left[\sum_{h=0}^{\infty} \psi_{h,j_1} \sin(\omega h)\right]^2 d\omega > 0.
$$

Since $\langle \cdot, \cdot\rangle^{freq}$ satisfies conjugate symmetry, linearity in the first argument, and positive-definiteness, it is an inner product operation mapping $\mathbb{R}^{\infty} \times \mathbb{R}^{\infty} \to \mathbb{R}$. $\square$

A.2 Theoretical Results for Gram Matrices

Notation. Let $\text{spec}(A)$ denote the spectrum of an arbitrary diagonalizable matrix A and $\bigoplus_{i\in I} A_i$ denote the direct sum of an ordered sequence of matrices $\{A_i\}_{i\in I}$. Denote the operation of the vector stack by $\oplus_{\text{stack}}$, that is, $a \oplus_{\text{stack}} b = (a', b')'$ for vectors a of size $n_a \times 1$ and b of size $n_b \times 1$. Similarly, denote the vector stack operation for an ordered sequence of vectors $\{a_i\}_{i\in\mathcal{I}}$ as $\bigoplus_{\text{stack}, i\in\mathcal{I}} a_i$. Denote the j-th column of the identity matrix I_n by δ_j^n. We use $\|\cdot\|$ to denote the Euclidean norm for vectors or the Frobenius norm for matrices and $\|\cdot\|_{\text{op}}$ to denote the operator norm for matrices.

Uniqueness of the Principal Eigenvector

We now discuss the claim that it is rare to have repeated eigenvealues in Ξ and thus rare for the max-share problem (9) to have a non-unique solution. In particular, the subset of Gram matrcies with repeated eigenvalues is of measure zero within the space of all Gram matrices under the standard Lebesgue measure, as is the subset of Gram matrices with non-unique principal eigenvectors.

Section 1.3.4 of Tao (2012) provides an argument for this claim more broadly for Hermitian matrices, leveraging on the differentiability of simple eigenvalues (similar to our proof of Lemma 5). Exercise 1.3.10 in Tao (2012) states that the space of Hermitian matrices of dimension $N \geq 2$ with at least one repeated eigenvalue has codimension 3 in the space of all Hermitian matrices.

The set of $N \times N$ Gram matrices with repeated eigenvalues is in the intersection of the set of all $N \times N$ Gram matrices and the set of all symmetric $N \times N$ matrices that have repeated eigenvalues. It can be shown that the set of symmetric matrices with repeated eigenvalues is a *real affine algebraic variety* in $\mathbb{R}^{N^*}$, where $N^* = N(N+1)/2$, and a proper subvariety of $\mathbb{R}^{N^*}$ (see, e.g., Lax, 1998). One can thus invoke a fundamental result in measure theory and algebraic geometry that a proper algebraic subvariety of $\mathbb{R}^{N^*}$ has Lebesgue measure zero in $\mathbb{R}^{N^*}$. Therefore, the set of $N \times N$ Gram matrices with repeated eigenvalues belongs to a set with Lebesgue measure zero.

Theorem 1

First, we prove two auxiliary Lemmas.

Lemma 3. *Let Ξ be a block diagonal Hermitian matrix with G diagonal blocks $\{\Xi_g\}_{g=1}^G$, each of size $n_g \times n_g$, denoted as $\Xi = \bigoplus_{g=1}^G \Xi_g$, then the following statements hold.*

(a) $\mathrm{spec}(\Xi) = \bigcup_{g=1}^G \mathrm{spec}(\Xi_g)$ *where* $\mathrm{spec}(\Xi_g) \subset \mathbb{R}$.

(b) If v_{g_j} is an $n_g \times 1$ eigenvector of the block Ξ_g corresponding to the eigenvalue $\lambda_{g_j} \in \mathrm{spec}(\Xi_g)$, then one can construct a $n \times 1$ block-sparse eigenvector v of Ξ corresponding to the same eigenvalue λ_{g_j} by padding $(n - n_g)$ zeros to v_g in all block components other than those corresponding to Ξ_g as $v = \mathbf{0}_{n_{g^-} \times 1} \oplus_{\text{stack}} v_{g_j} \oplus_{\text{stack}} \mathbf{0}_{n_{g^+} \times 1}$, where $n_0 = n_{G+1} \equiv 0$, $n_{g^-} = \sum_{g'=0}^{g-1} n_{g'}$, $n_{g^+} = \sum_{g''=g+1}^{G+1} n_{g''}$, and $n = \sum_{g=1}^G n_g = n_{g^-} + n_{g^+} + 1$.

(c) If there is a unique block Ξ_{g_0} such that its largest eigenvalue $\lambda_{\max}(\Xi_{g_0})$ is strictly larger than the largest eigenvalues of all other blocks Ξ_g for $g \neq g_0$, then the principal eigenvector v_0 of Ξ is block-sparse in the form of $v_0 = \mathbf{0}_{n_{g_0^-} \times 1} \oplus_{\text{stack}} v_{g_0} \oplus_{\text{stack}} \mathbf{0}_{n_{g_0^+} \times 1}$, where $n_0 = n_{G+1} \equiv 0$, $n_{g_0^-} = \sum_{g'=0}^{g_0-1} n_{g'}$, $n_{g_0^+} = \sum_{g''=g_0+1}^{G+1} n_{g''}$, and v_{g_0} is an eigenvector of Ξ_{g_0} corresponding to $\lambda_{\max}(\Xi_{g_0})$. Additionally, if $\lambda_{\max}(\Xi_{g_0})$ is a simple eigenvalue in $\mathrm{spec}(\Xi_{g_0})$ (of algebraic multiplicity 1), then v_0 is the unique principal eigenvector of Ξ.

Proof. Statement (a) follows because $\det(\Xi) = \prod_{g=1}^G \det(\Xi_g)$ for block diagonal matrices and the eigenvalues of a Hermitian matrix are real.

For (b), we verify $\Xi v = \lambda_{g_j} v$ as follows:

$$\begin{aligned}
\Xi v &= \left[\left(\bigoplus_{g'=1}^{g-1} \Xi_{g'}\right) \mathbf{0}_{n_{g-}\times 1}\right] \oplus_{\text{stack}} \left[\Xi_g v_{g_j}\right] \oplus_{\text{stack}} \left[\left(\bigoplus_{g''=g+1}^{G} \Xi_{g''}\right) \mathbf{0}_{n_{g+}\times 1}\right] \\
&= \left[\mathbf{0}_{n_{g-}\times 1}\right] \oplus_{\text{stack}} \left[\lambda_{g_j} v_{g_j}\right] \oplus_{\text{stack}} \left[\mathbf{0}_{n_{g+}\times 1}\right] \\
&= \lambda_{g_j} \left[\mathbf{0}_{n_{g-}\times 1} \oplus_{\text{stack}} v_{g_j} \oplus_{\text{stack}} \mathbf{0}_{n_{g+}\times 1}\right] = \lambda_{g_j} v.
\end{aligned}$$

For (c), it follows from (a) that $\lambda_{\max}(\Xi_{g_0}) \in \operatorname{spec}(\Xi)$ and $\lambda_{\max}(\Xi_{g_0}) = \lambda_{\max}(\Xi)$. Moreover, (b) implies that $v_0 = \mathbf{0}_{n_{g-}\times 1} \oplus_{\text{stack}} v_{g_0} \oplus_{\text{stack}} \mathbf{0}_{n_{g+}\times 1}$ is an eigenvector of Ξ corresponding to $\lambda_{\max}(\Xi_{g_0})$. The uniqueness of v_0 comes from the assumption that $\lambda_{\max}(\Xi_{g_0})$ is a simple eigenvalue in $\operatorname{spec}(\Xi_{g_0})$ and thus $\operatorname{spec}(\Xi)$. □

Lemma 4. *Let Ξ be an $n \times n$ Hermitian matrix. If δ_j^n is an eigenvector of Ξ for $1 \le j \le n$, then $\Xi = \Xi_{1:(j-1),1:(j-1)} \oplus \Xi_{j,j} \oplus \Xi_{(j+1):n,(j+1):n}$ and the eigenvalue corresponding to δ_j^n is $\Xi_{j,j}$.*

Proof. First, Ξ is Hermitian, so it only has real eigenvalues. Second, by the definition of an eigenvector, the jth column, $\Xi\delta_j^n$, of Ξ must be a multiple of δ_j^n, and so is the jth row of Ξ by symmetry. As such, $\Xi = \Xi_{1:(j-1),1:(j-1)} \oplus \Xi_{j,j} \oplus \Xi_{(j+1):n,(j+1):n}$ and $\Xi\delta_j^n = \Xi_{j,j}\delta_j^n$. □

Using Lemmas 3 and 4, we can prove Theorem 1.

Proof. Sufficiency ($\Rightarrow$): If $\Xi = \Xi_{1,1} \oplus \Xi_{2:N,2:N}$ and $\Xi_{1,1} > \lambda_{\max}(\Xi_{2:N,2:N})$, then $\lambda_{\max}(\Xi) = \Xi_{1,1}$ and has multiplicity 1 in $\operatorname{spec}(\Xi)$. By Lemma 3 (c), the principal eigenvector of Ξ is δ_1^N and is unique.

Necessity ($\Leftarrow$): If δ_1^N is an eigenvector of Ξ, then by Lemma 4, $\Xi = \Xi_{1,1} \oplus \Xi_{2:N,2:N}$ and the eigenvalue corresponding to δ_1^N is $\Xi_{1,1}$. Now suppose that $\Xi_{1,1} < \lambda_{\max}(\Xi_{2:N,2:N})$, then by Lemma 3 (c), the principal eigenvector of Ξ cannot be δ_1^N, which leads to a contradiction. Suppose that $\Xi_{1,1} = \lambda_{\max}(\Xi_{2:N,2:N})$, then by Lemma 3 (b), one can always construct another eigenvector that corresponds to $\lambda_{\max}(\Xi) = \Xi_{1,1} = \lambda_{\max}(\Xi_{2:N,2:N})$ and is orthogonal to δ_1^N, contradicting the uniqueness of δ_1^N as the principal eigenvector of Ξ. In summary, it has to be the case that the spectral gap of Ξ is strictly positive. □

Proposition 1

Proposition 1 is a special case of a standard result for the differential of eigenvalues and eigenvectors.[1] We reproduce it in a form that facilitates the interpretation of our results.

Lemma 5. *Let Ξ_0 be a real symmetric $n\times n$ matrix with a complete set of orthornormal eigenvectors $\{v_{0j}\}_{j=1}^n$ that span $\mathbb{R}^n$. Suppose v_{0i} is the eigenvector corresponding to a simple eigenvalue λ_{0i} of*

[1]See, for instance, Magnus and Neudecker (2019) for a textbook exposition for symmetric matrices, Aït-Sahalia and Xiu (2019) for its application in PCA analysis of high-frequency data, and Greenbaum, Li, and Overton (2020) for a comprehensive treatment for general square matrices.

Ξ_0 ($\Xi_0 v_{0i} = \lambda_{0i} v_{0i}$), for some $1 \le i \le n$. When Ξ_0 is perturbed by an infinitesimal and symmetric $\delta\Xi = o(\|\Xi_0\|)$, the following first order perturbation result holds for the new eigenpair (λ_i and v_i such that $\Xi v_i = \lambda_i v_i$) of the perturbed matrix, $\Xi = \Xi_0 + \delta\Xi$:

$$\lambda_i = \lambda_{0i} + v_{0i}' \delta\Xi v_{0i} + O\left(\|\delta\Xi\|^2\right),$$
$$v_i = v_{0i} + \sum_{\substack{j=1 \\ j \neq i}}^{n} \frac{v_{0j}' \delta\Xi v_{0i}}{\lambda_{0i} - \lambda_{0j}} v_{0j} + O\left(\|\delta\Xi\|^2\right).$$

Proof. First, by Theorem 8.9 in Magnus and Neudecker (2019), the functions $\lambda(\Xi)$ and $v(\Xi)$ are ∞ times differentiable in the neighborhood $N(\Xi_0) \subset \mathbb{R}^{n \times n}$ of Ξ_0. As such, it is valid to neglect higher order terms and focus on the first order approximation to the exact variation of λ_{0i} and v_{0i} in the event of an infinitesimal perturbation $\delta\Xi$ to Ξ_0, that is, $\delta\lambda_i$ and δv_i in $\lambda_i = \lambda_{0i} + \delta\lambda_i + O\left(\|\delta\Xi\|^2\right)$, and $v_i = v_{0i} + \delta v_i + O\left(\|\delta\Xi\|^2\right)$.

By construction, the perturbed matrix Ξ is symmetric and it is thus without loss of generality to normalize its eigenvectors $\{v_j\}_{j=1}^n$ such that

$$v_i' v_j = \delta_{ij}, \tag{A.1}$$

where δ_{ij} is the Kronecker delta. Note that, by neglecting higher order terms, (A.1) implies

$$\delta v_i' v_{0j} + v_{0i}' \delta v_j = 0, \text{ for } 1 \le i, j \le n. \tag{A.2}$$

By the definition of eigenvalue and eigenvector of Ξ, $\Xi v_i = \lambda_i v_i$. Substituting in $\Xi = \Xi_0 + \delta\Xi$ and the expansions of λ_i and v_i, and neglecting higher order terms, we obtain

$$\Xi_0 \delta v_i + \delta\Xi v_{0i} = \lambda_{0i} \delta v_i + \delta\lambda_i v_{0i}. \tag{A.3}$$

Left multiplying (A.3) with v_{0i}', we have

$$v_{0i}' \Xi_0 \delta v_i + v_{0i}' \delta\Xi v_{0i} = v_{0i}' \lambda_{0i} \delta v_i + \delta\lambda_i v_{0i}' v_{0i},$$

which, together with the facts that $\Xi_0 v_{0i} = \lambda_{0i} v_{0i}$ and $v_{0i}' v_{0i} = 1$, yields $\delta\lambda_i = v_{0i}' \delta\Xi v_{0i}$.

Now, note that $\{v_{0j}\}_{j=1}^n$ is orthonormal and spans $\mathbb{R}^n$ and $\delta v_i \in \mathbb{R}^n$, so $\delta v_i = \sum_{j=1}^n c_{ij} v_{0j}$, where $c_{ii} = v_{0i}' \delta v_i = 0$ from (A.2) taking $j = i$, and for $j \neq i$, c_{ij} is determined as follows. Left multiplying (A.3) with v_{0j}', we have

$$v_{0j}' \Xi_0 \delta v_i + v_{0j}' \delta\Xi v_{0i} = v_{0j}' \lambda_{0i} \delta v_i + \delta\lambda_i v_{0j}' v_{0i},$$

which, together with the facts that $\Xi_0 v_{0j} = \lambda_{0j} v_{0j}$ and $v_{0j}' v_{0i} = 0$, yields

$$c_{ij} = v_{0j}' \delta v_i = \frac{v_{0j}' \delta \Xi v_{0i}}{\lambda_{0i} - \lambda_{0j}}.$$

Therefore, we have obtained

$$\delta v_i = \sum_{\substack{j=1 \\ j \neq i}}^{n} \frac{v_{0j}' \delta \Xi v_{0i}}{\lambda_{0i} - \lambda_{0j}} v_{0j}.$$

□

We now specialize the results of Lemma 5 to our setting for Proposition 1.

Proof. $\Xi_0 \equiv \Xi_{1,1} \oplus \Xi_{2:N,2:N}$ is a real symmetric $N \times N$ matrix with the largest eigenvalue $\Xi_{1,1}$ and the principal eigenvector δ_1^N according to Theorem 1. Moreover, the perturbation $\delta\Xi = o(\|\Xi_0\|)$ by assumption. Thus, it is valid to specialize Lemma 5 to $\Xi = \Xi_0 + \delta\Xi$ by treating Ξ_0 as the unperturbed matrix and Ξ as the perturbed matrix.

By Lemma 3 (a), $\{\lambda_{0j}\}_{j=2}^N$ are also eigenvalues of Ξ_0. Also, define $v_{0j} = 0 \oplus_{\text{stack}} w_{0j}$ for $2 \leq j \leq N$. Then, $\delta_1^N \perp v_{0j}$ for any j and thus $\{\delta_1^N, \{v_{0j}\}_{j=2}^N\}$ constitutes a complete set of orthonormal eigenvectors of Ξ_0.

Denote the principal eigenvector of Ξ by v_1, which is unique by the assumption that $\lambda_{\max}(\Xi)$ is simple. By Lemma 5 and up to a first order approximation, $\lambda_{\max}(\Xi) - \lambda_{\max}(\Xi_0) = \left(\delta_1^N\right)' \delta\Xi\delta_1^N + O\left(\|\delta\Xi\|^2\right) = 0 + O\left(\|\delta\Xi\|^2\right) = O\left(\|\nu\|^2\right)$ by construction. Furthermore,

$$\begin{aligned} v_1 &= \delta_1^N + \sum_{j=2}^{N} \frac{v_{0j}' \delta\Xi\delta_1^N}{\Xi_{1,1} - \lambda_{0j}} v_{0j} + O\left(\|\nu\|^2\right) \\ &= \delta_1^N + \sum_{j=2}^{N} \frac{w_{0j}' \nu}{\Xi_{1,1} - \lambda_{0j}} v_{0j} + O\left(\|\nu\|^2\right) = 1 \oplus_{\text{stack}} \sum_{j=2}^{N} \frac{w_{0j}' \nu}{\Xi_{1,1} - \lambda_{0j}} w_{0j} + O\left(\|\nu\|^2\right) \end{aligned}$$

□

Proposition 2

Proposition 2 is an application of the Davis-Kahan $\sin\theta$ theorem (Davis and Kahan, 1970), which provides bounds on the distance between subspaces spanned by population eigenvectors and their sample analogs. These correspond, respectively, to eigenvectors of the unperturbed and perturbed matrices in our context. However, their bounds are usually in terms of eigengaps between certain population and sample eigenvalues. One of its variants (Yu et al., 2015) is then often used to derive such bounds in terms of the population eigenvalue separation condition and a smaller matrix norm (minimum of a scaled operator norm and the Frobenius norm). In the following auxiliary lemma, we specialize these bounds to study the change of the principal eigenvector in our context.

Lemma 6. *Let Ξ_0 be a real symmetric $n \times n$ matrix, with eigenvalues $\lambda_{01} > \lambda_{02} \geq \cdots \geq \lambda_{0n}$. Suppose Ξ_0 is perturbed by a symmetric $\delta\Xi$, resulting in another symmetric matrix Ξ, with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$. If v_{01} and v_1 are the (normalized) principal eigenvector of Ξ_0 and Ξ, respectively and denote the principal angel between them by $\Theta(v_{01}, v_1)$, then*

$$\sin\Theta(v_{01}, v_1) \leq \frac{2\left\|\delta\Xi\right\|_{\text{op}}}{\lambda_{01} - \lambda_{02}} = \frac{2\lambda_{\max}(\delta\Xi)}{\lambda_{01} - \lambda_{02}}.$$

Moreover, if $v_1' v_{01} \geq 0$, then

$$\|v_1 - v_{01}\| \leq \frac{2^{3/2}\left\|\delta\Xi\right\|_{\text{op}}}{\lambda_{01} - \lambda_{02}} = \frac{2^{3/2}\lambda_{\max}(\delta\Xi)}{\lambda_{01} - \lambda_{02}}.$$

Proof. This is a direct implication of Corollary 1 in Yu et al. (2015) for $j = 1$. The bound on the Euclidean distance between the principal eigenvectors is by the fact that

$$\|v_1 - v_{01}\|^2 = 2 - 2v_1'v_{01} = 2\left[1 - \cos\Theta(v_{01}, v_1)\right] \leq 2\left[1 - \cos^2\Theta(v_{01}, v_1)\right] = 2\sin^2\Theta(v_{01}, v_1).$$

□

Using the above Lemma 6, we now prove Proposition 2.

Proof. This is a direct application of Lemma 6 and Theorem 1 (which gives rise to the principal eigenvector and the eigengap for the unperturbed matrix $\Xi_0 \equiv \Xi_{1,1} \oplus \Xi_{2:N,2:N}$). It remains to calculate $\left\|\delta\Xi\right\|_{\text{op}}$. By the formula for the determinant of a block matrix, one has $\det(\lambda)\det\left(\lambda I_{N-1} - \frac{1}{\lambda}\nu\nu'\right) = 0$, whose largest nonzero root is $\|\nu\|$ and thus $\left\|\delta\Xi\right\|_{\text{op}} = \|\nu\|$. □

Theorem 2

Proof. By the linearity of the inner product $\langle\cdot,\cdot\rangle$, we have, for any j,

$$\langle\psi^*, \psi_j\rangle = \sum_{k=1}^{N}\theta_k\langle\psi_k, \psi_j\rangle = \lambda_{\max}(\Xi)\theta_j, \tag{A.4}$$

where the second equality follows from the Gramian structure of $\Xi_{j,k}$ and the j-th row of the eigenequation, $\Xi\theta = \lambda_{\max}(\Xi)\theta$. Summing (A.4) using weights θ_j and, again, by the linearity of the inner product, we have

$$\langle\psi^*, \psi^*\rangle = \sum_{k=1}^{N}\theta_k\langle\psi^*, \psi_j\rangle = \lambda_{\max}\sum_{k=1}^{N}\theta_k^2 = \lambda_{\max}(\Xi). \tag{A.5}$$

(A.4) and (A.5) jointly yield

$$\langle\psi^*, \psi_j\rangle = \theta_j\langle\psi^*, \psi^*\rangle = \theta_j\lambda_{\max}(\Xi).$$

Equation (17) follows from the linearity of the inner product.

For (18), notice that from (17) we have:

$$\left(\frac{\langle\psi^*,\psi_j\rangle}{\langle\psi^*,\psi^*\rangle}\right)^2 = \left(\sum_{j=2}^{N}\alpha_j\theta_j\right)^2 \leq \left(\sum_{j=2}^{N}\alpha_j^2\right)\left(\sum_{j=2}^{N}\theta_j^2\right) = 1-\theta_1^2.$$

The inequality follows from Cauchy-Schwarz (with equality when $\alpha_j = \theta_j/\sqrt{\sum_{j=2}^{N}\theta_j^2}$). The final equality follows from $\sum_{j=2}^{N}\alpha_j^2 = \sum_{j=1}^{N}\theta_j^2 = 1$. Rearranging terms yields equation (18). □

A.3 Special Cases

Rank One

Proof. Write $\Xi = vv'$. The unique non-zero eigenvalue is $\mathrm{Tr}(\Xi) = \sum_j v_j^2$. By the definition of the eigenvector, we have $\Xi\theta = \sum_j v_j^2\theta$, and thus $v_j\sum_{j'}v_{j'}\theta_{j'} = \theta_j\sum_{j'}v_{j'}^2$, or

$$\frac{\theta_j}{\sqrt{\Xi_{j,j}}} = \frac{\theta_j}{v_j} = \frac{\sum_{j'}v_{j'}\theta_{j'}}{\sum_{j'}v_{j'}^2}$$

for all j. □

Two Shocks

Proof. Suppose $N=2$. The eigenvector θ satisfies $(\Xi-\lambda I)\theta = 0$, which implies:

$$\lambda = \Xi_{1,1}+\Xi_{1,2}\frac{\theta_2}{\theta_1} = \Xi_{2,2}+\Xi_{1,2}\frac{\theta_1}{\theta_2}.$$

Multiplying throughout by θ_1/θ_2 and dividing by $\Xi_{1,2}$, we have:

$$\left(\frac{\theta_1}{\theta_2}\right)^2 - \vartheta\left(\frac{\theta_1}{\theta_2}\right) - 1 = 0,$$

where $\vartheta \equiv \frac{\Xi_{1,1}-\Xi_{2,2}}{\Xi_{1,2}}$. The quadratic has two solutions, with

$$\frac{\vartheta+\sqrt{\vartheta^2+4}}{2} \tag{A.6}$$

corresponding to the larger eigenvalue. □

A.4 Supply and Demand Example

Normalizing coefficient on the supply shock innovations, we can write the (38)-(39) as a VAR(1):

$$Y_t = GRG^{-1}Y_{t-1} + GS\varepsilon_t, \tag{A.7}$$

where

$$Y_t = \begin{bmatrix} q_t \\ p_t \end{bmatrix}, R = \begin{bmatrix} \rho^s & 0 \\ 0 & \rho^d \end{bmatrix}, S = \begin{bmatrix} \sigma^s & 0 \\ 0 & \sigma^d \end{bmatrix}, \text{ and } G = \frac{1}{\gamma^s + \gamma^d} \begin{bmatrix} \gamma^d & \gamma^s \\ -1 & 1 \end{bmatrix}.$$

The innovations ε_t are iid standard normal and ρ^d is the persistence of the demand shock. The true impulse response at horizon h of the ith variable to a one unit innovation in the jth shock is the (i, j) element of:

$$\Psi_h = GR^hS = \frac{1}{\gamma^s + \gamma^d} \begin{bmatrix} (\rho^s)^h\gamma^d\sigma^s & (\rho^d)^h\gamma^s\sigma^d \\ -(\rho^s)^h\sigma^s & (\rho^d)^h\sigma^d \end{bmatrix}. \tag{A.8}$$

Each of the elements of Ψ_h has the form $(\rho^{\mathsf{x}})^h\mathsf{q}$ where $\mathsf{x} \in \{s, d\}$ and q is the on-impact response, as in an AR(1).

References

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